# Feedback-enhanced quantum Chernoff discrimination

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**Abstract**

Classical adaptation cannot improve the Chernoff error exponent when each quantum copy is measured to completion on its first visit. We construct faithful qubit states for which retaining a quantum output for a second visit gives a strict improvement over every fixed single-copy measurement. The protocol uses partial measurements, local quantum storage, and classical communication to select the final readout. Its additive advantage grows without bound along a family of state pairs. A four-dimensional construction gives an exactly determined protocol exponent with matching finite-copy error bounds. We also prove that the optimal finite-round copywise LOCC exponent exists for every finite-dimensional pair. For orthogonally flagged pure conditional states, we derive explicit formulas for the measured and collective rates. LOCC and separable measurements attain the latter, and the ratio is bounded by a sharp factor of 3/2. Faithful four-dimensional approximations approach this ratio. For arbitrary faithful pairs, the standard fidelity comparison gives the upper bound two. None of the protocols requires a joint quantum operation on different copies.

## 1 Introduction

The quantum Chernoff theorem determines the optimal exponential decay of the error in discriminating repeated preparations of two quantum states [1, 2]. The optimal test may act jointly on all copies. When the copies remain in separate laboratories, measurements must instead be coordinated through classical communication. The achievable exponent then depends on the local operations and communication allowed by the protocol.

Hayashi proved that successive single-copy measurements have the same optimal Chernoff exponent as the best fixed single-copy POVM, provided each measurement is completed before the protocol moves to the next copy [3, Section V]. The measurement at each step may depend on the entire preceding record. Even this unrestricted classical adaptation leaves the exponent unchanged. General LOCC protocols allow a laboratory to retain a quantum output and measure it again after receiving further messages. Hayashi left open the corresponding question for two-way LOCC and separable measurements beyond the Stein setting [3, Section IX].

A partial measurement can produce a classical outcome while leaving a quantum state in local memory. The laboratory can share the outcome and choose its final measurement after receiving

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observations from the other copies. Local quantum storage thus permits later observations to influence the readout of an earlier input, using product inputs and classical communication alone.

Our four-dimensional construction separates the classical flag from the residual qubit. Reading all flags before measuring any retained qubit gives an error exponent above 0.68250256, whereas every fixed single-copy POVM has exponent below 0.677144. The construction was first reported in [4]. Matching upper and lower error bounds for every number of copies determine the protocol's exponent exactly.

A qubit already suffices for a strict separation. The first measurement generates a branch label and retains a quantum output on one branch. For a faithful family indexed by $\epsilon$, the two-pass protocol achieves an exponent exceeding the measured exponent by at least

$$\frac{1}{18}\log\frac{1}{\epsilon} - \log 3.$$

The additive advantage is therefore unbounded across state pairs, and local dimension two is minimal. For each pair, $\epsilon$ is fixed before the number of copies tends to infinity; the limit $\epsilon \downarrow 0$ is taken afterwards.

We also prove existence of the optimal finite-round copywise LOCC exponent for every finite-dimensional pair. A product inequality for optimal block errors bounds an arbitrary LOCC protocol tree and yields the limit. For orthogonally flagged pure conditional states, local attainability of binary pure-state discrimination [5] implies equality of the LOCC, separable, and collective rates. Explicit formulas for the measured and collective rates give a sharp upper bound of $3/2$ on their ratio. Faithful approximations attain the same limiting ratio. For general faithful pairs, standard fidelity inequalities give the upper bound two [6, 7, 8]. These ratio bounds coexist with an unbounded additive gap because the rates themselves diverge along the constructed families.

Calsamiglia, de Vicente, Muñoz-Tapia and Bagan studied mixed-qubit discrimination using finite-copy partial-transpose bounds and adaptive protocols [9]. Their numerical asymptotic fits do not provide a converse uniform over all faithful qubit pairs. Our family has very small positive eigenvalues and admits an analytic bound over all single-copy POVMs. The locality constraint here also differs from that of Matthews and Winter [10]: their two laboratories can jointly process the subsystems held on each side of a bipartition, whereas each laboratory here holds one whole copy. Adaptive channel-discrimination models such as [11] permit retained outputs to be measured collectively at the end. Our protocols never couple different input copies.

## 2 Copywise measurements and the LOCC exponent

All logarithms are natural. Let $\rho, \sigma$ be density operators on a finite-dimensional Hilbert space $\mathcal{H}$. Under the two equally likely hypotheses, $n$ laboratories receive $\rho^{\otimes n}$ or $\sigma^{\otimes n}$, with one tensor factor per laboratory. A binary test $\{T, I - T\}$ has error

$$p_{\mathrm{e}}(T; n) = \frac{1}{2}\left[\operatorname{Tr}\rho^{\otimes n}(I - T) + \operatorname{Tr}\sigma^{\otimes n}T\right]. \tag{1}$$

The class $\mathrm{LOCC}_n$ consists of tests implemented by a finite sequence of local instruments and classical messages. A laboratory may retain local quantum memory and be visited again. The number of operations may depend on $n$. There is no shared entangled resource, quantum communication, or operation on two input copies. We take infima over protocols, without assuming an optimal protocol is attained. Finite-outcome protocols suffice for these infima: in finite dimension a finite-depth general-outcome instrument tree can be approximated in its two input-state error probabilities by finite-outcome trees.

The class $\mathrm{SEP}_n$ consists of binary tests for which both effects are fully separable across the $n$ laboratories. In particular, $\mathrm{LOCC}_n \subseteq \mathrm{SEP}_n$. For either class $\mathcal{C}$, write

$$p_{\mathrm{e}}^{\mathcal{C}}(n) = \inf_{T \in \mathcal{C}_n} p_{\mathrm{e}}(T; n), \qquad \beta_n^{\mathcal{C}} = 2p_{\mathrm{e}}^{\mathcal{C}}(n),$$

and define

$$\underline{\xi}_{\mathcal{C}} = \liminf_{n\to\infty} -\frac{1}{n} \log p_{\mathrm{e}}^{\mathcal{C}}(n), \qquad \overline{\xi}_{\mathcal{C}} = \limsup_{n\to\infty} -\frac{1}{n} \log p_{\mathrm{e}}^{\mathcal{C}}(n). \tag{2}$$

For a finite-outcome single-copy POVM $M = \{M_i\}$, put $P_\tau^M(i) = \operatorname{Tr} \tau M_i$ and

$$C_{\mathrm{M}}(\rho, \sigma) = -\log \inf_{\substack{M \\ 0<s<1}} \sum_i P_\rho^M(i)^{1-s} P_\sigma^M(i)^s. \tag{3}$$

At non-full-rank states, endpoint values mean one-sided limits. For faithful states this agrees with optimization over $0 \le s \le 1$. The collective Chernoff exponent is

$$C_{\mathrm{Q}}(\rho, \sigma) = -\log \inf_{0<s<1} \operatorname{Tr} \rho^{1-s} \sigma^s. \tag{4}$$

Repeating a nearly optimal fixed POVM and applying the classical likelihood test achieves exponents arbitrarily close to $C_{\mathrm{M}}$. Hayashi's theorem identifies $C_{\mathrm{M}}$ with the optimum over single-visit adaptive measurements [3, Section V].

**Theorem 1** (Existence of the copywise LOCC exponent)**.** *For every finite-dimensional state pair and positive integers $n, m$,*

$$\beta_{n+m}^{\mathrm{LOCC}} \ge \beta_n^{\mathrm{LOCC}} \beta_m^{\mathrm{LOCC}}. \tag{5}$$

*Consequently, with the extended-real convention,*

$$\xi_{\mathrm{LOCC}} := \lim_{n\to\infty} -\frac{1}{n} \log p_{\mathrm{e}}^{\mathrm{LOCC}}(n) = \inf_{k\ge 1} -\frac{1}{k} \log \beta_k^{\mathrm{LOCC}}. \tag{6}$$

*Proof.* For nonnegative operators $A_i, B_i$ at $k$ distinct laboratories, define

$$V_k(\mathbf{A}, \mathbf{B}) = \inf_{\{E, I-E\} \in \mathrm{LOCC}_k} \left\{ \operatorname{Tr}\left[ \left( \bigotimes_i A_i \right) (I - E) \right] + \operatorname{Tr}\left[ \left( \bigotimes_i B_i \right) E \right] \right\}.$$

If a local instrument $\{\mathcal{N}_x\}$ acts at laboratory $j$, then

$$V_k(\mathbf{A}, \mathbf{B}) \le \sum_x V_k(A_1, B_1; \ldots; \mathcal{N}_x(A_j), \mathcal{N}_x(B_j); \ldots; A_k, B_k). \tag{7}$$

Indeed, applying that instrument and following a near-optimal continuation for each outcome is an admissible protocol. For finitely many outcomes, the sum of the approximation errors can be made arbitrarily small.

Split $n + m$ laboratories into blocks $U, W$ of sizes $n, m$. At every history of an arbitrary finite LOCC tree, the two unnormalized conditional inputs remain products across laboratories. Define

$$H = V_n(\mathbf{A}_U, \mathbf{B}_U) V_m(\mathbf{A}_W, \mathbf{B}_W).$$

When an instrument acts in $U$, the second factor is independent of its outcome, so (7) gives $H(\text{parent}) \le \sum_x H(\text{child } x)$. The same holds for an instrument in $W$.

At a terminal history put

$$a = \prod_{i \in U} \operatorname{Tr} A_i, \quad c = \prod_{i \in U} \operatorname{Tr} B_i, \quad b = \prod_{i \in W} \operatorname{Tr} A_i, \quad d = \prod_{i \in W} \operatorname{Tr} B_i.$$

A constant decision is allowed in the definition of $V_k$, hence

$$H \le \min(a, c) \min(b, d) \le \min(ab, cd).$$

The actual terminal decision has cost either $ab$ or $cd$. Summing from the leaves to the root therefore bounds the error overlap of every such protocol below by $H(\text{root}) = \beta_n^{\text{LOCC}} \beta_m^{\text{LOCC}}$. Local ancillas and retained quantum memory are included in the current local operators; shared classical randomness can be included in the transcript. Taking the infimum proves (5).

If the states are not orthogonal, $\beta_1^{\text{LOCC}} > 0$, and (5) implies $\beta_n^{\text{LOCC}} \ge (\beta_1^{\text{LOCC}})^n > 0$. The sequence $a_n = -\log \beta_n^{\text{LOCC}}$ is subadditive. Fekete's lemma gives $\lim_n a_n/n = \inf_n a_n/n$, and the factor two relating $\beta_n$ to $p_{\text{e}}(n)$ has no effect on the limit. Orthogonal states can already be distinguished at one laboratory, giving the infinite-exponent case. □

The product argument uses the sequential structure of an LOCC instrument tree. It does not establish a corresponding product inequality or ordinary limit for general separable tests.

**Proposition 2** (Fidelity comparison)**.** *For distinct faithful finite-dimensional states,*

$$C_{\text{M}} \le \xi_{\text{LOCC}} \le \underline{\xi}_{\text{SEP}} \le \overline{\xi}_{\text{SEP}} \le C_{\text{Q}} \le 2C_{\text{M}}. \tag{8}$$

*Proof.* Let $F = (\operatorname{Tr} |\sqrt{\rho}\sqrt{\sigma}|)^2$ be the squared fidelity. A fidelity-achieving one-copy POVM has classical Bhattacharyya coefficient $\sqrt{F}$ [7], so $C_{\text{M}} \ge -\frac{1}{2} \log F$. The Fuchs–van de Graaf inequality and the Helstrom formula give

$$\beta_n^{\text{ALL}} = 1 - \tfrac{1}{2} \|\rho^{\otimes n} - \sigma^{\otimes n}\|_1 \ge 1 - \sqrt{1 - F^n} \ge \tfrac{1}{2} F^n.$$

The quantum Chernoff theorem yields $C_{\text{Q}} \le -\log F \le 2C_{\text{M}}$ [1, 2, 8]. The remaining inequalities follow from repeated single-copy measurements, Theorem 1, and class inclusion. This standard comparison also appears in the study of local Chernoff distinguishability [6]. □

## 3 A four-dimensional separation

Each input consists of two orthogonal qubit blocks. The block label can be read without disturbing the conditional qubit, so the final measurement basis can depend on the labels from all copies.

Set

$$\begin{gathered} \varepsilon = 2^{-16}, \qquad a = 1 - \varepsilon, \qquad p = \frac{4}{5}, \qquad A = \begin{pmatrix} a & 0 \\ 0 & \varepsilon \end{pmatrix}, \\ H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad B = HAH, \\ \rho = pA \oplus (1-p)B, \qquad \sigma = (1-p)B \oplus pA. \end{gathered} \tag{9}$$

For $0 < u < 1$, let $h_2(u) = -u \log u - (1-u) \log(1-u)$.

**Theorem 3.** *The states in (9) are faithful and noncommuting. Their measured Chernoff exponent obeys*

$$C_{\mathrm{M}}(\rho,\sigma) < \log\frac{5}{4} + \frac{454}{1000} < 0.677144. \tag{10}$$

*There is a two-pass finite-round LOCC protocol with error $p_{\mathrm{e}}^{\mathrm{tp}}(n)$ satisfying, for every integer $n \geq 1$,*

$$\frac{e^{-d-nE}}{4(n+1)^2} \leq p_{\mathrm{e}}^{\mathrm{tp}}(n) \leq \frac{n+1}{2}e^{-nE}, \tag{11}$$

*where*

$$d = \log\frac{a}{\varepsilon} = \log 65535, \qquad u = \frac{\log(2a)}{d}, \qquad E = \log\frac{5}{4} + \log 2 - h_2(u). \tag{12}$$

*In particular, the protocol has an ordinary error exponent equal to $E$, and*

$$\underline{\xi}_{\mathrm{SEP}} \geq \xi_{\mathrm{LOCC}} \geq E > C_{\mathrm{M}}(\rho,\sigma) + \frac{3}{1000}. \tag{13}$$

*An outward-rounded evaluation of (12) gives $0.68250256 < E < 0.68250257$.*

The theorem supplies an explicit counterexample to the universal equality $\xi_{\mathrm{LOCC}} = \underline{\xi}_{\mathrm{SEP}} = \overline{\xi}_{\mathrm{SEP}} = C_{\mathrm{M}}$ posed in the QIQCOP entry [12]. The value $E$ is the exact exponent of the displayed protocol; it is a lower bound on the optimal-class exponents.

### 3.1 Uniform bound on the measured exponent

The following coefficient estimate applies to arbitrary finite POVMs, including measurements that do not read the direct-sum flag first.

**Lemma 4.** *Let $X, Y > 0$ be density matrices, $1/2 \leq \alpha < 1$, and $\beta = (1-\alpha)/\alpha$. Every finite POVM $\{M_i\}$ satisfies*

$$\sum_i (\operatorname{Tr} XM_i)^{\alpha} (\operatorname{Tr} YM_i)^{1-\alpha} \geq \left(\operatorname{Tr} XY^{\beta}\right)^{\alpha}. \tag{14}$$

*Proof.* Omit zero effects and write $x_i = \operatorname{Tr} XM_i$, $y_i = \operatorname{Tr} YM_i$. Scalar minimization gives

$$\sum_i x_i^{\alpha} y_i^{1-\alpha} = \inf_{t_i>0} \left[\alpha \sum_i x_i t_i^{-\beta} + (1-\alpha)\sum_i y_i t_i\right].$$

The minimizing values are $t_i = (x_i/y_i)^{\alpha}$. For a fixed choice, let $G = \sum_i t_i M_i > 0$ and $v = \operatorname{Tr} YG > 0$. Operator Jensen's inequality for $z^{-\beta}$ gives

$$\sum_i t_i^{-\beta} M_i \geq G^{-\beta}.$$

Also $G^{1/2} Y G^{1/2} \leq vI$, so $Y \leq vG^{-1}$. The Loewner–Heinz inequality, valid here because $0 \leq \beta \leq 1$, yields $Y^{\beta} \leq v^{\beta} G^{-\beta}$. Thus, with $c = \operatorname{Tr} XY^{\beta}$, the expression being minimized is at least

$$\alpha c v^{-\beta} + (1-\alpha)v \geq c^{\alpha}.$$

The last scalar minimum occurs at $v = c^{\alpha}$. This proves (14). Appendix A recalls the operator inequalities used in this argument. □

Let $\Pi_0, \Pi_1$ project onto the two summands of $\mathcal{H} = \mathbb{C}^2 \oplus \mathbb{C}^2$. Refining an arbitrary POVM effect $M_i$ into $\Pi_j M_i \Pi_j$ preserves its total probabilities on the block-diagonal states. For $0 < s < 1$, scalar Hölder gives

$$\sum_j x_j^{1-s} y_j^s \le \Big(\sum_j x_j\Big)^{1-s} \Big(\sum_j y_j\Big)^s.$$

Consequently, refinement can only decrease the Chernoff coefficient. The effects restricted to either summand form a qubit POVM.

For $\alpha = \max(s, 1-s)$ and $\beta = (1-\alpha)/\alpha$, apply Lemma 4 in both blocks, interchanging the two hypotheses when needed. Since $H$ interchanges $A$ and $B$,

$$\operatorname{Tr} AB^\beta = \operatorname{Tr} BA^\beta = \frac{a^\beta + \varepsilon^\beta}{2}.$$

Every original POVM coefficient is therefore at least

$$\frac{4}{5} \cosh((2\alpha - 1)\log 2) \left(\frac{a^\beta + \varepsilon^\beta}{2}\right)^\alpha. \tag{15}$$

**Lemma 5.** *For the states (9), the logarithm of (15) is strictly greater than* $\log(4/5) - 454/1000$, *uniformly for* $1/2 \le \alpha < 1$.

*Proof.* Write $L = \log 2$. Since $a^\beta + \varepsilon^\beta \ge a^\beta(1 + \varepsilon^\beta)$, the logarithm in question is at least

$$\log \frac{4}{5} + g(\alpha) + (1-\alpha)\log a, \tag{16}$$

where

$$g(\alpha) = \log\cosh((2\alpha-1)L) - \alpha L + \alpha \log\left(1 + e^{-16L(1-\alpha)/\alpha}\right).$$

The last term is a convex perspective of $z \mapsto \log(1 + e^z)$, evaluated on an affine function of $\alpha$. The first term has second derivative

$$4L^2 \operatorname{sech}^2((2\alpha-1)L) \ge \frac{64}{25} L^2 > \frac{6}{5}$$

on $[1/2, 1]$. Thus $g''(\alpha) > 6/5$.

At $\alpha = 3/4$, put $x = 2^{-16/3}$. Direct differentiation gives

$$g(3/4) = \log 3 - \frac{9}{4}L + \frac{3}{4}\log(1+x),$$
$$g'(3/4) = -\frac{1}{3}L + \log(1+x) + \frac{64L}{3}\frac{x}{1+x}.$$

The rational enclosures

$$\frac{6931}{10000} < L < \frac{6932}{10000}, \quad \log 3 > \frac{10986}{10000}, \quad \frac{248}{10000} < x < \frac{249}{10000} \tag{17}$$

together with $x - x^2/2 \le \log(1+x) \le x$ imply

$$g(3/4) > -\frac{443}{1000}, \qquad 0 < g'(3/4) < \frac{77}{500}.$$

Strong convexity and completion of the square now give, throughout $[1/2, 1]$,

$$g(\alpha) > -\frac{443}{1000} - \frac{(77/500)^2}{12/5}.$$

Finally, $\log a \geq -\varepsilon/(1-\varepsilon) = -1/65535$ and $1-\alpha \leq 1/2$. Therefore

$$g(\alpha) + (1-\alpha)\log a > -\delta, \qquad \delta = \frac{443}{1000} + \frac{(77/500)^2}{12/5} + \frac{1}{131070} < \frac{454}{1000}. \tag{18}$$

The margins are rational; Appendix B gives a direct certificate for (17) and the displayed comparisons. □

Equations (15)–(18) give a common coefficient lower bound $(4/5)e^{-\delta}$ for every POVM and every interior Chernoff parameter. At $s = 0, 1$ the coefficient equals one because the states are faithful. Taking the infimum over both the measurement and the parameter proves (10).

### 3.2 The two-pass protocol

Identify each direct summand with a qubit, and let $V_j : \mathcal{H} \to \mathbb{C}^2$ extract summand $j$. The first local instrument consists of

$$\mathcal{I}_j(X) = V_j X V_j^*, \qquad j = 0, 1.$$

Its sum is trace preserving. The classical outcome $j$ is the flag, and the quantum output is the retained qubit. Their laws are

| | $j=0$ | $j=1$ |
|---|---|---|
| $\rho$ | $p,\ A$ | $1-p,\ B$ |
| $\sigma$ | $1-p,\ B$ | $p,\ A.$ |

First apply this instrument to every input copy and communicate all flags. Let $k$ be the number of zero flags. If $k \geq n/2$, each party measures in the eigenbasis of its conditional state under $\rho$: the $A$ basis for flag zero and the $B$ basis for flag one. If $k < n/2$, it uses the corresponding $\sigma$ eigenbasis. In either basis, order the outcomes with the larger eigenvalue first. Finish with the optimal classical likelihood decision using the complete transcript.

Put $P = (a, \varepsilon)$ and $Q = (1/2, 1/2)$. Measuring $A$ in the $B$ basis, or $B$ in the $A$ basis, gives $Q$; measuring either in its own basis gives $P$. Hence, conditional on any particular flag string, the second-pass laws are $P^{\otimes n}, Q^{\otimes n}$ in the majority-zero branch, and $Q^{\otimes n}, P^{\otimes n}$ in the other branch. The weights of a string with $k$ zeros are

$$w_k = p^k(1-p)^{n-k}, \qquad v_k = (1-p)^k p^{n-k}. \tag{19}$$

This is an exact finite-round LOCC implementation. Both passes may use a fixed visiting order. Each party retains one qubit between the passes and is measured once in the second pass. A pass denotes a visit to every party, rather than a single individual LOCC operation. Every terminal transcript effect is a product of local positive effects. Grouping transcripts by the final decision makes both binary effects fully separable.

### 3.3 Exact error exponent of the protocol

For the residual classical pair, define for $0 \leq t \leq 1$

$$r(t) = \sum_i P_i^t Q_i^{1-t} = 2^{t-1}(a^t + \varepsilon^t).$$

Its logarithm is strictly convex. At its stationary point $t_*$, the tilted small-outcome probability equals $u = \log(2a)/d$, since the tilted mean of $\log(P_i/Q_i)$ vanishes. Thus

$$t_* = \frac{\log((1-u)/u)}{d}, \qquad -\log r(t_*) = \log 2 - h_2(u) =: C_{\mathrm{cl}}(P, Q). \tag{20}$$

In particular $0 < t_* < 1/2$. To check the interval explicitly, $u < 1/2$ follows from $4a\varepsilon < 1$. Also $\log a \geq -1/65535$ and $d < 16 \log 2$ imply $u > 1/32$. At $t = 0$ the tilted small-outcome probability is $1/2$, whereas at $t = 1/2$ it is $\sqrt{\varepsilon}/(\sqrt{a} + \sqrt{\varepsilon}) < 1/128$. It decreases strictly with $t$, proving the claim. The identity for $C_{\mathrm{cl}}$ follows from $(1-u)\log a + u \log \varepsilon = -\log 2$.

### 3.3.1 Upper bound on the error

Fix a flag string in the branch $k \geq n/2$. The classical overlap of its two weighted conditional laws is at most

$$\sum_z \min\{w_k P^{\otimes n}(z), v_k Q^{\otimes n}(z)\} \leq w_k^{t_*} v_k^{1-t_*} r(t_*)^n.$$

For $x = k/n$ the flag-type weight satisfies

$$\binom{n}{k} w_k^{t_*} v_k^{1-t_*} \leq e^{n f(x)},$$

where

$$f(x) = h_2(x) + t_*[x \log p + (1-x)\log(1-p)] + (1-t_*)[x \log(1-p) + (1-x)\log p].$$

Here we used $\binom{n}{k} \leq e^{n h_2(k/n)}$, including the endpoints by continuity. For $x \geq 1/2$,

$$f'(x) = \log \frac{1-x}{x} + (2t_* - 1)\log 4 < 0,$$

and $f(1/2) = \log(4/5)$. In the branch $k < n/2$, reverse the Chernoff powers; the type function is $f(1-x)$ and gives the same bound. Summing the $n+1$ types, including the boundary type once, and inserting the equal-prior factor proves

$$p_{\mathrm{e}}^{\mathrm{tp}}(n) \leq \frac{n+1}{2} \left(\frac{4}{5} r(t_*)\right)^n = \frac{n+1}{2} e^{-nE}.$$

### 3.3.2 A matching lower bound

Retain only flag strings with $k = \lfloor n/2 \rfloor$. If $n$ is even, their two weights equal $(2/5)^n$. If $n$ is odd, the weights are $(2/5)^n/2$ and $2(2/5)^n$. Thus for either parity their weighted overlap is at least $(2/5)^n/2$ times

$$S_n = \sum_z \min\{P^{\otimes n}(z), Q^{\otimes n}(z)\}.$$

The central binomial coefficient is at least $2^n/(n+1)$, so

$$p_{\mathrm{e}}^{\mathrm{tp}}(n) \geq \frac{(4/5)^n}{4(n+1)} S_n. \tag{21}$$

This bound also covers the branch in which the two conditional laws are reversed.

Let $R = (1-u, u)$ be the tilted law at $t_*$. A mode $\ell = \lfloor (n+1)u \rfloor$ of its binomial count satisfies $|\ell - nu| \leq 1$, and its type probability is at least $1/(n+1)$. On that type the log-likelihood ratio is

$$L_z = \log \frac{P^{\otimes n}(z)}{Q^{\otimes n}(z)} = d(nu - \ell), \qquad |L_z| \leq d.$$

The exact tilted identities give

$$P^{\otimes n}(z) = r(t_*)^n R^{\otimes n}(z) e^{(1-t_*)L_z},$$
$$Q^{\otimes n}(z) = r(t_*)^n R^{\otimes n}(z) e^{-t_* L_z}.$$

Their minimum is at least $e^{-d} r(t_*)^n R^{\otimes n}(z)$. Summing this one type proves $S_n \geq e^{-d} r(t_*)^n/(n+1)$. Equation (21) yields the lower bound in (11). The polynomial factors in the two error bounds disappear after taking $-n^{-1}\log$, establishing the exact exponent $E$.

### 3.3.3 Strict separation

The residual exponent has the elementary lower bound

$$\begin{aligned} C_{\mathrm{cl}}(P,Q) &\geq -\log r(1/4) \\ &= \frac{3}{4}\log 2 - \log(a^{1/4} + 1/16) \\ &> \frac{3}{4}\log 2 - \log(17/16) > \frac{3}{4}\frac{6931}{10000} - \frac{1}{16} > \frac{457}{1000}. \end{aligned}$$

Combining this with (10) gives $E - C_{\mathrm{M}} > 3/1000$. Optimal LOCC and separable errors cannot exceed the displayed protocol's error, which proves (13). Finally, both states have trace one and minimum eigenvalue $\varepsilon/5 = 1/327680$. The off-diagonal entry of $[A,B]$ is $(a-\varepsilon)^2/2 \neq 0$, so $[\rho,\sigma] \neq 0$. This completes the proof of Theorem 3.

## 4 Faithful qubit states with an unbounded exponent gap

A partial qubit measurement can generate a classical branch label while retaining a nontrivial quantum output in one branch. A separate flag register is therefore unnecessary.

**Theorem 6** (Faithful qubit separation)**.** *For $0 < \epsilon < 1/2$, set $\delta = \epsilon^2$, $L = \log(1/\epsilon)$, and*

$$\begin{aligned} |\phi\rangle &= \sqrt{\epsilon}\,|0\rangle + \sqrt{1-\epsilon}\,|1\rangle, \\ \rho_\epsilon &= (1-\delta)|0\rangle\langle 0| + \delta|1\rangle\langle 1|, \\ \sigma_\epsilon &= (1-\delta)|\phi\rangle\langle\phi| + \delta(I - |\phi\rangle\langle\phi|). \end{aligned} \tag{22}$$

*These states are faithful and noncommuting. They obey*

$$C_{\mathrm{M}}(\rho_\epsilon,\sigma_\epsilon) \leq \frac{2}{3}L, \qquad \xi_{\mathrm{LOCC}}(\rho_\epsilon,\sigma_\epsilon) \geq \frac{13}{18}L - \log 3. \tag{23}$$

*The second bound is achieved by a two-pass protocol whose overlap satisfies*

$$\beta_n^{\mathrm{qubit}} \leq (3\epsilon^{13/18})^n \qquad (n \geq 1). \tag{24}$$

*In particular, $\epsilon = 2^{-36}$ certifies $\xi_{\mathrm{LOCC}} - C_{\mathrm{M}} \geq \log(4/3) > 0$. Local dimension two is minimal for a strict separation.*

### 4.1 Bound on the measured exponent

The measured-rate bound follows from a pure-state moment inequality, also used in Section 5.

**Lemma 7.** *For normalized pure states $|\alpha\rangle, |\gamma\rangle$ with squared overlap $v$, the outcome laws $P_x, Q_x$ of any POVM satisfy*

$$\sum_x P_x^{1-s} Q_x^s \geq v^{\max(s,1-s)} \qquad (0 < s < 1). \tag{25}$$

*For each fixed $s$, an appropriate binary projective measurement attains equality.*

*Proof.* Positivity of each effect and Cauchy–Schwarz give

$$\sqrt{v} = |\langle\alpha|\gamma\rangle| \leq \sum_x |\langle\alpha|M_x|\gamma\rangle| \leq \sum_x \sqrt{P_x Q_x} =: B.$$

For $s \leq 1/2$, write

$$\sqrt{P_x Q_x} = (P_x^{1-s} Q_x^s)^{1/[2(1-s)]} Q_x^{(1-2s)/[2(1-s)]}.$$

Hölder's inequality implies $B \leq (\sum_x P_x^{1-s} Q_x^s)^{1/[2(1-s)]}$, proving the lower bound $v^{1-s}$. Interchanging the hypotheses proves the other half. Projection onto $|\gamma\rangle$ attains $v^{1-s}$ for $s \leq 1/2$, and projection onto $|\alpha\rangle$ attains $v^s$ for $s \geq 1/2$. □

The common depolarizing channel $\mathcal{D}(X) = (1-2\delta)X + \delta \operatorname{Tr}(X) I$ maps the pure pair $|0\rangle, |\phi\rangle$ to (22). Pulling a POVM back through $\mathcal{D}$ and using Lemma 7 gives $Q_M(s) := \sum_x p_x^{1-s} q_x^s \geq \epsilon^{\max(s,1-s)}$. Moreover $\delta I \leq \rho_\epsilon, \sigma_\epsilon \leq (1-\delta) I$. With $r = \delta/(1-\delta)$, every outcome satisfies $p_x \geq r q_x$ and $q_x \geq r p_x$, and therefore $Q_M(s) \geq r^{\min(s,1-s)}$. Since $\log(1/r) \leq 2L$,

$$-\log Q_M(s) \leq L \min\{\max(s, 1-s), 2\min(s, 1-s)\} \leq \frac{2}{3} L. \tag{26}$$

This proves the first part of (23) uniformly in the POVM and the Chernoff parameter.

The leading measured rate is also determined. In the computational basis the laws are

$$(1-\delta, \delta), \qquad (z, 1-z), \qquad z = \epsilon + \delta(1-2\epsilon).$$

At $s = 2/3$, their moment is at most $\epsilon^{2/3}[(1+\epsilon)^{2/3} + 1]$. Consequently

$$C_{\mathrm{M}}(\rho_\epsilon, \sigma_\epsilon) = \frac{2}{3} L + O(1) \qquad (\epsilon \downarrow 0). \tag{27}$$

### 4.2 First-pass instrument and conditional readouts

Put $t = \sqrt{\epsilon}$, $P = |\phi\rangle\langle\phi|$, and use the first-pass Kraus operators

$$K_1 = P + \sqrt{t}\,(I - P), \qquad K_0 = \sqrt{1-t}\,(I - P). \tag{28}$$

They satisfy $K_0^\dagger K_0 + K_1^\dagger K_1 = I$. Branch zero has a rank-one output, so its residual quantum state carries no further information about the hypothesis. In branch one retain the qubit and put

$$w = t + (1-t)\epsilon, \qquad |\psi\rangle = K_1|0\rangle/\sqrt{w}.$$

After all branch labels have been announced, choose one of two final measurements for every retained qubit: $L = \{P, I-P\}$ or $R = \{|\psi\rangle\langle\psi|, I - |\psi\rangle\langle\psi|\}$. The choice will depend on the full branch string. A classical likelihood decision then uses all recorded outcomes.

The unnormalized outcome laws in branch zero are

$$p_0 = (1-t)(1-z), \qquad q_0 = (1-t)\delta. \tag{29}$$

In branch one, with the projector outcome listed first,

$$P_{1L} = (z, t(1-z)), \qquad Q_{1L} = (1-\delta, t\delta), \tag{30}$$

$$P_{1R} = (u, \bar{u}), \qquad Q_{1R} = (v, \bar{v}), \tag{31}$$

where

$$u = (1-\delta)w + \delta\frac{(1-t)^2\epsilon(1-\epsilon)}{w}, \qquad \bar{u} = \frac{\delta t}{w},$$

$$v = \frac{(1-\delta)\epsilon + \delta t(w-\epsilon)}{w}, \qquad \bar{v} = 1 - \delta + t\delta - v.$$

Direct evaluation gives, for example, $|\langle\psi|\phi\rangle|^2 = \epsilon/w$ and $\bar{u} = \delta\det(K_1^\dagger K_1)/w = \delta t/w$. These probabilities include the first-pass branch weights.

The probabilities satisfy

$$u \le w, \qquad v \le \epsilon/w, \qquad \bar{u} \le \delta, \qquad \bar{v} \le 1, \qquad w \ge t. \tag{32}$$

For the first inequality, $(1-t)^2\epsilon(1-\epsilon) \le \epsilon = t^2 \le w^2$. For the second, $t(w-\epsilon) = \epsilon(1-\epsilon) \le \epsilon$. The remaining inequalities follow directly from the probabilities above.

### 4.3 Two-pass error bound

Let $A_{jT}(s)$ be the order-$s$ moment of the unnormalized laws in branch $j$ with readout $T$. For branch zero it is $p_0^{1-s}q_0^s$, independently of $T$. Equations (29)–(32) give the bounds

$$A_{0L}(1/6) \le \epsilon^{1/3}, \qquad A_{1L}(1/6) \le 2\epsilon^{5/6}, \tag{33}$$

$$A_{0R}(3/4) \le \epsilon^{3/2}, \qquad A_{1R}(3/4) \le 2\epsilon^{1/2}. \tag{34}$$

For the nontrivial $L$ bound,

$$A_{1L}(1/6) \le \epsilon^{5/6}\left[(z/\epsilon)^{5/6} + (1-z)^{5/6}\right] \le 2\epsilon^{5/6},$$

because $(z/\epsilon) + (1-z) = 2 - 3\epsilon^2 + 2\epsilon^3 \le 2$ and $x^{5/6}$ is concave. For $R$,

$$A_{1R}(3/4) \le w^{1/4}(\epsilon/w)^{3/4} + \delta^{1/4} \le 2\epsilon^{1/2}.$$

For a fixed first-pass string with $m$ retained branches, using $L$ bounds the terminal overlap by

$$X = (\epsilon^{1/3})^{n-m}(2\epsilon^{5/6})^m,$$

whereas using $R$ bounds it by

$$Y = (\epsilon^{3/2})^{n-m}(2\epsilon^{1/2})^m.$$

The protocol compares the two bounds and selects the readout using the shared first-pass string. Using $\min(X,Y) \le X^{2/3}Y^{1/3}$,

$$\beta_n^{\text{qubit}} \le \sum_{m=0}^{n}\binom{n}{m}X^{2/3}Y^{1/3} = \sum_{m=0}^{n}\binom{n}{m}2^m\epsilon^{13n/18} = (3\epsilon^{13/18})^n.$$

The exponent 13/18 is the same for both branch types: $(2/3)(1/3) + (1/3)(3/2) = 13/18$ and $(2/3)(5/6) + (1/3)(1/2) = 13/18$. Theorem 1 now gives the second bound in (23). For $\epsilon = 2^{-36}$, $L/18 - \log 3 = \log(4/3)$. The states have positive eigenvalues $\delta, 1-\delta$, and their eigenbases differ nontrivially for $0 < \epsilon < 1/2$. They are therefore faithful and noncommuting. One-dimensional normalized states are identical, proving the minimal-dimension assertion.

The construction gives a lower bound on the optimal rate, not its exact value. It also implies $\liminf_{\epsilon \downarrow 0} \xi_{\mathrm{LOCC}}/C_{\mathrm{M}} \geq 13/12$. Although the noise tends to zero along this family, no state is singular at a fixed $\epsilon > 0$. The asymptotic number of copies is always taken first.

**Proposition 8** (Stability under faithful perturbations)**.** *Suppose $e^{-\eta}\rho \leq \rho' \leq e^{\eta}\rho$ and $e^{-\eta}\sigma \leq \sigma' \leq e^{\eta}\sigma$ for some $\eta \geq 0$. Then*

$$|\xi_{\mathrm{LOCC}}(\rho', \sigma') - \xi_{\mathrm{LOCC}}(\rho, \sigma)| \leq \eta, \qquad |C_{\mathrm{M}}(\rho', \sigma') - C_{\mathrm{M}}(\rho, \sigma)| \leq \eta.$$

*A gap larger than $2\eta$ therefore persists under such a perturbation.*

*Proof.* Tensoring the operator inequalities and tracing against the two positive decision effects gives $e^{-n\eta}\beta_n^{\mathcal{C}}(\rho, \sigma) \leq \beta_n^{\mathcal{C}}(\rho', \sigma') \leq e^{n\eta}\beta_n^{\mathcal{C}}(\rho, \sigma)$ for any fixed test class. Each one-copy outcome probability changes by a factor between $e^{-\eta}$ and $e^{\eta}$, so the same holds for every measured moment. Take logarithms, optimize, and use Theorem 1. □

For a faithful pair these relative operator inequalities hold in a sufficiently small norm neighborhood. The separation therefore holds on an open set of state pairs, although the argument does not estimate the size of that set.

## 5 Exact rates for orthogonally flagged pure conditional states

For pure conditional states, reading the flags reduces the problem to binary pure-state discrimination. The theorem of Virmani, Sacchi, Plenio and Markham [5] then gives a local measurement attaining the global error on every flag string. Equality of the restricted and collective rates follows immediately. The proof below uses an elementary two-pass test within a factor two of the finite-copy optimum, which suffices to determine the exponent. An explicit measured-rate formula gives the sharp ratio bound.

Let a finite flag $f$ index mutually orthogonal blocks, and consider

$$\rho = \bigoplus_f p_f |\psi_f\rangle\langle\psi_f|, \qquad \sigma = \bigoplus_f q_f |\phi_f\rangle\langle\phi_f|, \qquad v_f = |\langle\psi_f|\phi_f\rangle|^2 > 0, \tag{35}$$

where $p_f, q_f > 0$ and $\sum_f p_f = \sum_f q_f = 1$. Define

$$G(s) = \sum_f p_f^{1-s} q_f^s v_f^{\max(s,1-s)}, \qquad F(s) = \sum_f p_f^{1-s} q_f^s v_f.$$

Here $F(s)$ denotes a Chernoff moment, not the squared fidelity used in Proposition 2.

**Theorem 9** (Exact flagged-state law)**.** *For the pair (35),*

$$C_{\mathrm{M}} = -\log \inf_{0<s<1} G(s), \qquad \xi_{\mathrm{LOCC}} = \xi_{\mathrm{SEP}} = C_{\mathrm{Q}} = -\log \inf_{0<s<1} F(s) \leq \frac{3}{2} C_{\mathrm{M}}. \tag{36}$$

*Both restricted exponents exist as ordinary limits. The factor 3/2 is asymptotically sharp within this class.*

*Proof.* Refining a one-copy POVM by the orthogonal flags can only decrease its order-$s$ moment, by scalar Hölder. Within each flag, Lemma 7 gives the least possible moment, and the appropriate projective measurement attains it. Thus the least one-copy moment is $G(s)$. Pure-state powers also give $\operatorname{Tr}\rho^{1-s}\sigma^s = F(s)$.

For $n$ copies, read every flag. Conditional on a flag string, the unnormalized hypotheses are $a|\Psi\rangle\langle\Psi|$ and $b|\Phi\rangle\langle\Phi|$, where both vectors are products and $V = |\langle\Psi|\Phi\rangle|^2 = \prod_i v_{f_i}$. If $a \geq b$, each laboratory projects onto $|\psi_{f_i}\rangle$ and the protocol declares $\rho$ exactly when all projections pass. Its overlap cost is $bV$. If $a < b$, reverse the roles, giving cost $aV$.

The unrestricted optimal overlap for these two weighted pure states is

$$\beta^{\mathrm{ALL}}(a,b,V) = \frac{a+b-\sqrt{(a+b)^2-4abV}}{2} = \frac{2abV}{a+b+\sqrt{(a+b)^2-4abV}} \geq \frac{1}{2}\min(a,b)V.$$

The displayed local test therefore costs at most twice the optimum on every flag string. The strings occupy orthogonal blocks, so summing them yields

$$\beta_n^{\mathrm{ALL}} \leq \beta_n^{\mathrm{SEP}} \leq \beta_n^{\mathrm{LOCC}} \leq 2\beta_n^{\mathrm{ALL}} \qquad (n \geq 1). \tag{37}$$

The quantum Chernoff theorem proves equality and ordinary convergence of all three exponents.

For the factor bound, first fix $0 < s \leq 1/2$ and set $s' = 2s/3$. Since $1 - s' \geq 2/3$ and $0 < v_f \leq 1$,

$$G(s') \leq \sum_f \left[p_f^{1-s} q_f^s v_f\right]^{2/3} p_f^{1/3} \leq F(s)^{2/3}.$$

For $s \geq 1/2$, use $s' = (2s+1)/3$ and the same Hölder argument with $q_f$ in place of $p_f$. Thus $\inf_{s'} G(s') \leq (\inf_s F(s))^{2/3}$, which gives $C_{\mathrm{Q}} \leq \frac{3}{2} C_{\mathrm{M}}$. □

### 5.1 Sharpness of the factor 3/2

Choose $0 < v < 1$, $p = 1/(1+v)$, $q = v/(1+v)$, and pure qubit vectors with $|\langle\psi|\phi\rangle|^2 = v$. Set

$$\rho_v^{(0)} = \operatorname{diag}(p,q) \otimes |\psi\rangle\langle\psi|, \qquad \sigma_v^{(0)} = \operatorname{diag}(q,p) \otimes |\phi\rangle\langle\phi|. \tag{38}$$

For $s \leq 1/2$, $G(s) = (v + v^{2-2s})/(1+v)$; for $s \geq 1/2$, $G(s) = G(1-s)$. Its infimum is $v$, attained in the one-sided endpoint limit. On the other hand, $F(s) = v(v^s + v^{1-s})/(1+v)$ is minimized at $s = 1/2$. Consequently

$$C_{\mathrm{M}} = \log(1/v), \qquad \xi_{\mathrm{LOCC}} = \xi_{\mathrm{SEP}} = C_{\mathrm{Q}} = -\log\frac{2v^{3/2}}{1+v}. \tag{39}$$

The ratio approaches $3/2$ as $v \downarrow 0$.

### 5.2 Faithful approximations

The pure conditional states in (38) can be replaced by strictly positive qubit states while retaining the leading rates. For $T > 1$ set

$$v = e^{-T}, \qquad \delta = e^{-T^3}, \qquad a = 1 - \delta/2, \qquad b = \delta/2, \tag{40}$$

and let

$$A_\delta = a|\psi\rangle\langle\psi| + b(I - |\psi\rangle\langle\psi|), \qquad B_\delta = a|\phi\rangle\langle\phi| + b(I - |\phi\rangle\langle\phi|).$$

Define $\rho_{v,\delta} = \operatorname{diag}(p,q) \otimes A_\delta$ and $\sigma_{v,\delta} = \operatorname{diag}(q,p) \otimes B_\delta$, with $p, q$ as above. These four-dimensional states are faithful and noncommuting for every fixed $T > 1$.

**Proposition 10.** *For the family (40), taking the number of copies to infinity at fixed $T$ before letting $T \to \infty$,*

$$C_{\mathrm{M}} = T + o(1), \qquad \xi_{\mathrm{LOCC}} = \frac{3}{2}T - \log 2 + o(1). \tag{41}$$

*In particular, $\xi_{\mathrm{LOCC}}/C_{\mathrm{M}} \to 3/2$ along faithful state pairs.*

*Proof.* The states are obtained from (38) by the same depolarizing channel on the conditional qubit, so measured-divergence data processing gives $C_{\mathrm{M}} \le T$. In the basis containing $|\phi\rangle$, the qubit laws are $(u, 1-u)$ and $(a, b)$, where $u = av + b(1-v) \le v + b$. The basis containing $|\psi\rangle$ reverses the roles. For $0 < \eta < 1/2$, put

$$m_{v,\delta}(\eta) = a^{\eta}u^{1-\eta} + b^{\eta}(1-u)^{1-\eta} \le (v+b)^{1-\eta} + b^{\eta}.$$

After reading all flags, let $A_z, B_z$ be their unnormalized likelihoods. Choosing the final basis with the smaller of the two moment bounds gives overlap at most

$$m_{v,\delta}(\eta)^n \min\{A_z^{1-\eta}B_z^{\eta}, A_z^{\eta}B_z^{1-\eta}\} \le m_{v,\delta}(\eta)^n \sqrt{A_z B_z}.$$

Summing the flag strings gives

$$\beta_n^{\mathrm{LOCC}} \le \left[\frac{2\sqrt{v}}{1+v} m_{v,\delta}(\eta)\right]^n. \tag{42}$$

For sufficiently large $T$, choose $\eta = T^{-3/2}$. The two terms in the moment bound have different scales:

$$(v+b)^{1-\eta} = e^{-T+T^{-1/2}}(1+o(1)), \qquad b^{\eta} = \exp(-T^{3/2} - T^{-3/2}\log 2) = o(e^{-T}).$$

Thus $m_{v,\delta}(\eta) \le e^{-T+T^{-1/2}}(1+o(1))$, and (42) implies $\xi_{\mathrm{LOCC}} \ge \frac{3}{2}T - \log 2 - o(1)$.

For the converse, the quantum Chernoff moment factorizes into flag and qubit moments. Both are symmetric and log-convex about $s = 1/2$, so

$$\inf_s \operatorname{Tr} \rho_{v,\delta}^{1-s}\sigma_{v,\delta}^{s} = \frac{2\sqrt{v}}{1+v}\left[v + 2(1-v)\sqrt{ab}\right] = 2e^{-3T/2}(1+o(1)).$$

Hence $\xi_{\mathrm{LOCC}} \le C_{\mathrm{Q}} = \frac{3}{2}T - \log 2 + o(1)$.

Finally, reading the flag and measuring in the $|\phi\rangle$ basis at order $s = 1/4$ gives the one-copy moment

$$\frac{v^{1/4} + v^{3/4}}{1+v}\left[a^{1/4}u^{3/4} + b^{1/4}(1-u)^{3/4}\right] = e^{-T}(1+o(1)).$$

Together with $C_{\mathrm{M}} \le T$, this proves the measured-rate assertion. □

The upper bound $3/2$ in Theorem 9 concerns the pure-conditional flagged class. Proposition 10 gives a sequence of faithful examples approaching that ratio; it does not extend the $3/2$ upper bound to all faithful pairs.

## 6 Discussion

Local quantum memory allows the final measurement on one copy to depend on observations of other copies. The four-dimensional protocol stores a conditional qubit after reading an orthogonal flag. The qubit protocol retains a quantum output after a partial measurement. Both exploit information that remains quantum during the exchange of classical outcomes.

The error bounds account for every first-pass record. In the four-dimensional protocol, flag strings close to a tie determine the rate: their classical cost contributes $\log(5/4)$, and the retained qubits contribute $C_{\mathrm{cl}}(P,Q)$. For the qubit family, the two readouts bound different compositions of the first-pass string. Combining their moments gives the same power $\epsilon^{13/18}$ for every composition and hence a uniform bound on the total error.

A pass visits each relevant input once; it can contain many local operations and messages. The separation from single-visit measurements therefore concerns the value of revisiting a quantum input. Whether further visits yield a strict hierarchy of exponents remains open.

The optimal exponent of the faithful qubit family is still unknown. For general faithful pairs, the sharp amplification factor lies between $3/2$ and two. Theorem 1 establishes the LOCC limit, but convergence of the optimal SEP rate and a possible LOCC–SEP exponent separation remain unresolved in general. The pure-conditional flagged class has equal LOCC, SEP, and collective rates; faithful approximations preserve the limiting ratio to the measured exponent without establishing equality of all three rates at fixed nonzero noise.

## Author contributions and AI assistance

The explicit four-dimensional construction and its exact protocol exponent appeared in Zhang's manuscript of 16 October 2026 [4]. Li's discussion draft of 21 October 2026 supplied the LOCC product-value argument, the faithful qubit construction, and the flagged-state analysis [13]. The faithful-approximation proof uses the noise scale in (40).

Computational checks supplement the analytic proofs. For the four-dimensional example, these include rational certificates and an independent integer evaluation of the transcript error. The qubit formulas and faithful-approximation estimates were checked with high-precision arithmetic, including direct matrix calculations and finite-transcript error evaluations.

Yuxuan Zhang set research directions, designed the harness coordinating model tasks, tools, memory, and checking, supplied physical and mathematical intuitions and proof skeletons, and guided the development and revision of the research ideas. In that work, AI models developed detailed candidate arguments and derivations, searched the literature, wrote and ran computational checks, and drafted and revised the manuscript under his supervision. Mao-Sheng Li supplied the complementary mathematical draft described above. AI assistance was also used to consolidate and revise the present manuscript. K-Dense's scientific-writing skill [14] assisted manuscript organization and evidence tracking.

## A The operator inequalities in Lemma 4

The two operator inequalities follow from inverse order and a resolvent integral. If $Wv = (M_i^{1/2}v)_i$ and $T = \mathrm{diag}(t_i I)$, then $W^*W = I$ and $W^*TW = G$. Compression of the positive block matrix

$$\begin{pmatrix} T & I \\ I & T^{-1} \end{pmatrix} \geq 0$$

by $\mathrm{diag}(W, W)$, followed by a Schur complement, gives $\sum_i t_i^{-1} M_i \geq G^{-1}$. Applying the same argument to $T + \lambda I$ gives $\sum_i (t_i + \lambda)^{-1} M_i \geq (G + \lambda I)^{-1}$. For $0 < \beta < 1$, integrate this inequality against the positive measure $\frac{\sin \pi\beta}{\pi} \lambda^{-\beta}\, d\lambda$ and use

$$z^{-\beta} = \frac{\sin \pi\beta}{\pi} \int_0^\infty \frac{\lambda^{-\beta}}{z + \lambda}\, d\lambda.$$

This proves the required Jensen inequality. The case $\beta = 1$ is the inverse inequality itself. Similarly, the representation

$$z^\beta = \frac{\sin \pi\beta}{\pi} \int_0^\infty \lambda^{\beta-1} \frac{z}{z+\lambda}\, d\lambda$$

and $z(z+\lambda I)^{-1} = I - \lambda(z+\lambda I)^{-1}$ show that $X \le Y$ implies $X^\beta \le Y^\beta$. The endpoint cases are immediate. All matrices here are positive definite, so the inverses and the convergent integrals are well defined.

## B Rational bounds and numerical evaluation

The logarithm brackets in (17) can be checked without floating-point arithmetic. For $0 < w < 1$,

$$0 < \log \frac{1+w}{1-w} - 2 \sum_{j=0}^{N-1} \frac{w^{2j+1}}{2j+1} \le \frac{2w^{2N+1}}{(2N+1)(1-w^2)}. \tag{43}$$

Use $(w, N) = (1/3, 12)$ for $\log 2$ and $(1/2, 16)$ for $\log 3$. The bounds on $x$ follow by cubing the rational endpoints and comparing with $2^{-16}$. Explicit bounds for the two midpoint quantities are

$$\begin{aligned}
g(3/4) &> \frac{10986}{10000} - \frac{9}{4}\frac{6932}{10000} + \frac{3}{4}\left(\frac{248}{10000} - \frac{1}{2}\left(\frac{249}{10000}\right)^2\right) > -\frac{443}{1000}, \\
g'(3/4) &> -\frac{1}{3}\frac{6932}{10000} + \frac{248}{10000} - \frac{1}{2}\left(\frac{249}{10000}\right)^2 + \frac{64}{3}\frac{6931}{10000}\frac{248/10000}{1+249/10000} > 0, \\
g'(3/4) &< -\frac{1}{3}\frac{6931}{10000} + \frac{249}{10000} + \frac{64}{3}\frac{6932}{10000}\frac{249/10000}{1+248/10000} < \frac{77}{500}.
\end{aligned}$$

Together with $\frac{64}{25}(6931/10000)^2 > 6/5$ and (18), these are a finite rational certificate for the measured bound. The analytic separation follows already from $C_{\mathrm{cl}} > 457/1000$.

For the more precise decimal value of $E$, the supplied evaluator applies (43) after power-of-two range reduction, with rational interval arithmetic throughout. It encloses $u$, evaluates $h_2(u)$, and checks the two endpoints stated in Theorem 3. In particular, combining the interval with (10) certifies $E - C_{\mathrm{M}} > 0.005359$. This sharper decimal is supplementary to the rational gap $3/1000$.

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